

Lecture 6

Diagonalization, Cayley–Hamilton, Functions of Operators

Throughout, the core uses finite-dimensional V over the stated $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, $T \in \mathcal{L}(V)$, same-field functions $f : \text{spec}(T) \rightarrow \mathbb{F}$, and $\chi_T(\lambda) = \det(\lambda I - T)$ is the characteristic polynomial. In the quantum preview, $|\nu\rangle$ is simply Dirac notation for the vector ν ; bras are introduced in Lecture 7. Tags mark the flavour; a $\star$ marks a harder one.

Core tutorial route

Work **6.1, 6.2, 6.7, 6.11, 6.14, 6.15, 6.17, and 6.26** in that order (about 80–100 minutes before discussion). This eight-problem route requires a complete diagonalization, a defective diagnosis, powers, an authentic Cayley–Hamilton reduction and proof, a finite matrix exponential, dynamical transfer, and spectral mapping.

Additional practice: 6.3–6.6, 6.8–6.10, 6.12–6.13, 6.16, 6.21–6.22. **Bridge:** 6.19, 6.23–6.24 (convergence and matrix dynamics). **Extension:** 6.18, 6.20, 6.25, 6.27–6.28 (minimal-polynomial recognition, Jordan/defective exponentials, ladder operators, and a stronger determinant identity). Extension tasks do not count toward core progress.

Diagonalization

Exercise 6.1 (computational) Diagonalize $A = \begin{pmatrix} 3 & 2 \\ 1 & 2 \end{pmatrix}$: find P and D with $P^{-1}AP = D$.

Answer. Eigenvalues 1, 4; $P = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$, $D = \text{diag}(4, 1)$.

Exercise 6.2 (computational) Decide whether $A = \begin{pmatrix} 5 & 1 \\ 0 & 5 \end{pmatrix}$ is diagonalizable, with reasons.

Answer. Not diagonalizable: the only eigenvalue 5 has a one-dimensional eigenspace.

Exercise 6.3 (computational) For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ (eigenvalues 3, 1, eigenvectors $(1, 1)$, $(1, -1)$) compute A^n for general n .

Answer. $A^n = \frac{1}{2} \begin{pmatrix} 3^{n+1} & 3^n - 1 \\ 3^n - 1 & 3^{n+1} \end{pmatrix}$.

Exercise 6.4 (computational) Diagonalize $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ over $\mathbb{R}$: find P and D with $P^{-1}AP = D$.

Answer. Eigenvalues 3, -1 ; $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $D = \text{diag}(3, -1)$.

Exercise 6.5 (computational) Diagonalize $A = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}$ over $\mathbb{C}$.

Answer. Eigenvalues $2 \pm i$; $P = \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}$, $D = \text{diag}(2 + i, 2 - i)$.

Exercise 6.6 (computational) Decide whether $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ is diagonalizable, with reasons.

Answer. Not diagonalizable: the only eigenvalue 2 has a one-dimensional eigenspace.

Exercise 6.7 (computational) For $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ compute A^n for general $n \geq 0$.

Answer. $A^n = \frac{1}{2} \begin{pmatrix} 3^n + (-1)^n & 3^n - (-1)^n \\ 3^n - (-1)^n & 3^n + (-1)^n \end{pmatrix}$.

Exercise 6.8 (proof) Suppose $T \in \mathcal{L}(V)$ is diagonalizable and has only one eigenvalue λ . Prove that $T = \lambda I$.

Cayley–Hamilton

Exercise 6.9 (computational) Use the Cayley–Hamilton theorem to express A^{-1} as a polynomial in A for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

Answer. $A^{-1} = \frac{1}{2}(A - 5I) = \begin{pmatrix} \frac{-2}{2} & \frac{1}{2} \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$.

Exercise 6.10 (computational) Use the Cayley–Hamilton theorem to express A^{-1} as a polynomial in A for $A = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$.

Answer. $A^{-1} = \frac{1}{10}(7I - A) = \begin{pmatrix} \frac{2}{5} & -\frac{1}{10} \\ -\frac{1}{5} & \frac{3}{10} \end{pmatrix}$.

Exercise 6.11 (computational) For $A = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$, use Cayley–Hamilton to write A^4 in the form $\alpha A + \beta I$, and give the resulting matrix.

Answer. $A^4 = 15A - 14I = \begin{pmatrix} -14 & 15 \\ -30 & 31 \end{pmatrix}$.

Exercise 6.12 (computational) For $A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 1 & 2 \end{pmatrix}$ compute χ_A and use Cayley–Hamilton to express A^{-1} as a polynomial in A ; give A^{-1} explicitly.

Answer. $\chi_A(\lambda) = \lambda^3 - 7\lambda^2 + 16\lambda - 12$; $A^{-1} = \frac{1}{12}(A^2 - 7A + 16I) = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ -\frac{1}{6} & \frac{1}{3} & 0 \\ \frac{1}{12} & -\frac{1}{6} & \frac{1}{2} \end{pmatrix}$.

Exercise 6.13 (proof) Suppose $T \in \mathcal{L}(V)$ is invertible on a complex space of dimension n . Prove that T^{-1} is a polynomial in T .

Exercise 6.14 (proof) Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. By direct computation show that $A^2 - (\text{tr } A)A + (\det A)I = 0$, thereby verifying Cayley–Hamilton for every 2×2 matrix.

Functions of operators

Exercise 6.15 (computational) For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ compute e^{tA} .

Answer. $e^{tA} = \frac{1}{2} \begin{pmatrix} e^{3t} + e^t & e^{3t} - e^t \\ e^{3t} - e^t & e^{3t} + e^t \end{pmatrix}$.

Exercise 6.16 (proof) For an arbitrary $T \in \mathcal{L}(V)$, prove directly that for any polynomials p, q , the operators $p(T)$ and $q(T)$ commute. Do not assume diagonalizability.

Exercise 6.17 (computational) Let $A = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$. Compute e^{tA} and describe the solution curves of $\dot{x} = Ax$.

Answer. $e^{tA} = \begin{pmatrix} \cos 2t & -\sin 2t \\ \sin 2t & \cos 2t \end{pmatrix}$; trajectories are circles at angular frequency 2.

Exercise 6.18 (★) **Extension (minimal-polynomial recognition)**. Suppose T is diagonalizable with eigenvalues in $\{0, 1\}$ (a projection-like operator). Prove that $T^2 = T$.

Hint. Diagonalize T ; the claim reduces to checking $\lambda^2 = \lambda$ on each eigenvalue.

Exercise 6.19 (★) Suppose A is diagonalizable and every eigenvalue of A satisfies $|\lambda| < 1$. Prove that $A^n \rightarrow 0$ as $n \rightarrow \infty$ (entrywise).

Hint. Diagonalize and examine D^n entrywise; the conjugating matrices P, P^{-1} are fixed.

Exercise 6.20 (computational) **Extension (Jordan–Chevalley and defective exponential).**

Write $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ as $A = D + N$ with D diagonalizable, N nilpotent, $DN = ND$, and use it to compute e^{tA} .

Answer. $D = 2I$, $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$; $e^{tA} = e^{2t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$.

Exercise 6.21 (computational) For $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ compute e^{tA} .

Answer. $e^{tA} = \frac{1}{2} \begin{pmatrix} e^{3t} + e^{-t} & e^{3t} - e^{-t} \\ e^{3t} - e^{-t} & e^{3t} + e^{-t} \end{pmatrix}$.

Exercise 6.22 (computational) Find the positive square root $\sqrt{A}$ of $A = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ (the one with positive eigenvalues), so that $(\sqrt{A})^2 = A$.

Answer. $\sqrt{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

Exercise 6.23 (computational) Compute e^{tA} for $A = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}$.

Answer. $e^{tA} = \begin{pmatrix} \cos 3t & \sin 3t \\ -\sin 3t & \cos 3t \end{pmatrix}$.

Exercise 6.24 (computational) Solve $\dot{x} = Ax$ with $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ and $x(0) = (2, 0)$.

Answer. $x(t) = (e^{3t} + e^{-t}, e^{3t} - e^{-t})$.

Exercise 6.25 (computational) **Extension (defective exponential).** Compute e^{tA} for $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.

Answer. $e^{tA} = \begin{pmatrix} e^{2t} & te^{2t} & 0 \\ 0 & e^{2t} & 0 \\ 0 & 0 & e^{3t} \end{pmatrix}$.

Exercise 6.26 (computational) Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, with $\text{spec}(A) = \{1, 3\}$. Using spectral mapping, find $\text{spec}(B)$ for $B = A^3 - 2A + I$.

Answer. $\text{spec}(B) = \{0, 22\}$ (with the same eigenvectors $(1, -1), (1, 1)$).

Exercise 6.27 (★) **Extension (finite ladder analogy).** On the spin-1 representation the raising operator acts by $J_+|-1\rangle = \sqrt{2}|0\rangle$, $J_+|0\rangle = \sqrt{2}|1\rangle$, $J_+|1\rangle = 0$. Write its matrix in the basis $(|1\rangle, |0\rangle, |-1\rangle)$ and show J_+ is nilpotent.

Hint. The columns of the matrix are the images of the basis kets. A strictly upper-triangular matrix is nilpotent — compute its powers.

Exercise 6.28 (proof) **Extension.** Let A be diagonalizable. Prove that $\det(e^A) = e^{\text{tr}A}$.